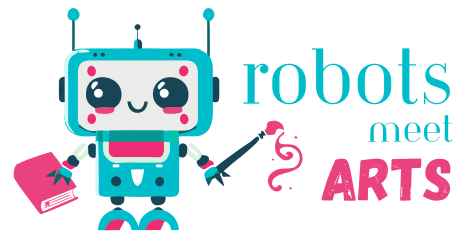




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FR01-KA220-SCH-
000151881

You as a Work of Art

**Subject:**

Art & design

Content:

Create your own art face filter

**Computational practices:
choose what fits**

- **Decomposing**
- Pattern recognition
- Abstraction
- **Algorithms**
- **Problem solving**

Unplugged/**plugged:**

**Preparation time:**

30 min

**Age:**

8-12 years

**Duration:**

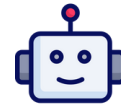
150min-180min

**Location:**

Classroom or art room

**Group distribution:**

In groups of two

**Difficulty****After this lesson, the children will be able to...**

- break down a complex task (face filter creation) into smaller, manageable steps.
- create a sequence of commands using block-based code in Scratch.
- use loops to repeat actions within their program.
- reflect on their coding process and explain their programming choices.
- design and sketch character features that reflect their artistic vision.
- digitize their artwork and prepare it for interactive use.
- explore and analyze artwork using digital tools



Materials needed:

- 1 laptop/tablet with camera for every group of 2 children
- 1 blank template for every group of 2 children
- 4 inspiration sheets for every group of 2 children

Craft materials:

- pencil, eraser and sharpener
- colored pencils
- markers
- paint and brushes
- paper
- colored paper
- scissors
- glue
- crayons
- ...

Helpful links:

Scratch & AI: FaceSensing | Tutorial:

<https://www.youtube.com/watch?v=uLFh9Fvuj1k&t=290s>

Scratch and AI inspiration cards:

<https://resources.scratch.mit.edu/www/cards/en/facesensing-cards.pdf>

Face Detection Games in Scratch with AI

<https://www.youtube.com/watch?v=ynISkhphQWg>

Preparation and teacher notes

Tutorial of the tool Google culture and arts:

<https://youtube.com/shorts/qojH2qx9M-w?si=djy5LkZX0Ek6JExY>

Warm up 20'

In pairs, students open Google Arts & Culture on their tablets and use the Art Projector to view artwork in an interactive way. Students then choose a piece of art they would like to continue working on.

Sketch phase:

Each group is given a blank main template on which they design a face filter to incorporate themselves into the artwork. They draw AR elements such as a hat, nose, glasses or mask to creatively fit themselves into the painting.

Build up 40'

Students can choose from the following 3 options to get their AR elements or you as a teacher can choose to offer only 1 option.

1: Craft material.

Have students draw their chosen AR elements each on a separate A4 sheet. Encourage creativity by giving them the freedom to use different materials, such as: (paint, marker, pencil, colored paper,...).

2: Digital tool (Tate.org.uk)

Have students work out the AR elements using the digital tool Tate. This easy-to-use digital drawing application offers a wide range of materials that students can use creatively, such as: (paint, pencil, marker,...)

3: Scratch drawing tool (Scratch lab).

Have students create their AR elements using the drawing tool in Scratch. This tool is easy to use and provides basic shapes and drawing options that allow students to express their creativity. The background does not need to be removed afterwards in this tool.

Preparation and teacher notes

You can also split the lesson into 2 parts. Part 1 making the AR elements with craft materials and part 2 programming the AR elements.

Watch this video for more information how to delete a background in Scratch Lab:
<https://www.youtube.com/watch?v=Za0eJwV8jvk>

Craft materials 60'

Collect on a table various craft materials that students can use. Possible materials you can offer are:

- Paint
- Crayons
- markers
- Colored paper , scissors
- Glue

Note: offers 2D materials only.



Digitalization of the AR elements.

Scan the AR elements by using a scanner. Then make sure each group has access to their scanned AR element when they start creating the face filter.

Upload the AR element into Scratch lab and remove the background.

Criteria in the AR element creation process.

- Each AR element they want to integrate into their face filter should be drawn on a separate A4 paper.
- Make sure that around the AR element no stains or background is drawn, this is going to cause difficulties when removing the background.
- If stains are still present around the AR element, cut out the AR element.
- Note that AR elements have a clear edge. Outlining the AR elements can help with this.
- Color your AR element completely so that no white paper is visible.

Preparation and teacher notes

Surf to the following website:

<https://www.tate.org.uk/kids/games-quizzes/tate-paint>

Digital tool 30'

Open the digital tool on a tablet. Then click on the "Let's Create" button and select the yellow canvas. This makes it easier to remove the background because the white canvas contains speckles.



The students can use different crafting materials present in the tool Tate.

When they are satisfied with their AR element the children click on the button with the green check and say yes to the question "Have you finished?". Finally, they click the green download button. This way they will find the AR element in the downloads of the tablet. Give it a clear name such as Hat.

Upload the AR element into Scratch lab and remove the background.

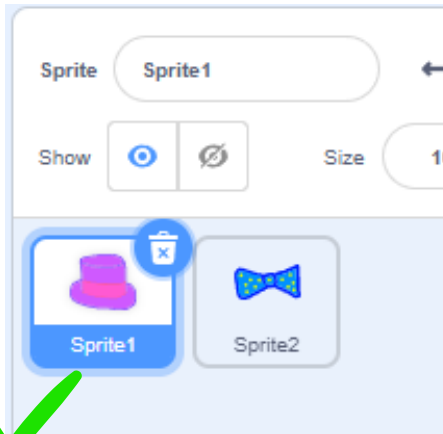
Criteria in the creation process of the AR elements with the digital tool Tate.

- Each AR element must be created on a separate canvas. No multiple AR element should be drawn together on a canvas.
- Make sure that no spots or background is drawn around the AR element, this is going to cause difficulties when removing the background.
- Make sure that AR elements have a clear border. Outlining the AR elements can help with this.
- Color your AR element completely so that no yellow is visible from the canvas.

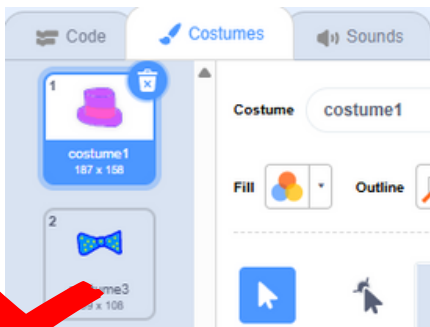
Preparation and teacher notes

How to use the scratch paint editor:

https://www.youtube.com/watch?v=9C_wblf4FIE



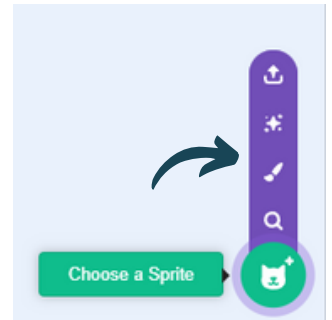
Correct, each AR element has an individual "sprite".



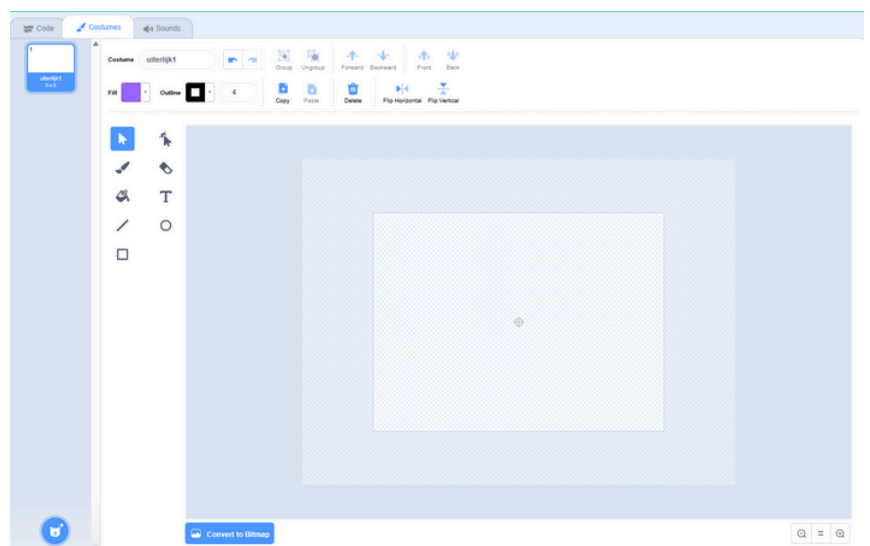
Incorrect, this way you cannot program the AR elements separately.

Scratch Paint editor (Scratch lab) 30'

Open the website <https://lab.scratch.mit.edu/> en selecteer "Face sensing" Klik vervolgens op deze knop en selecteer het penseel.



Then there will open this screen, on this screen you can use the different tools present in Scratch. On this drawing page, they can draw and modify their AR element.



Criteria in the AR element creation process.

- Each AR element is drawn as a separate "sprite." They do this by always clicking the right button as shown above.
- Color your AR element completely so that no gaps can be seen in your design.
- Place your AR element in the center of your sheet. You can move it by using the arrow in the upper left corner. Make sure here that all your elements are selected when dragging. You can do this by drawing a rectangle around your AR element. This way you can be sure that you have selected everything.

Preparation and teacher notes

When starting this section, make sure each group has access to the AR elements they created themselves and can use them in this part.

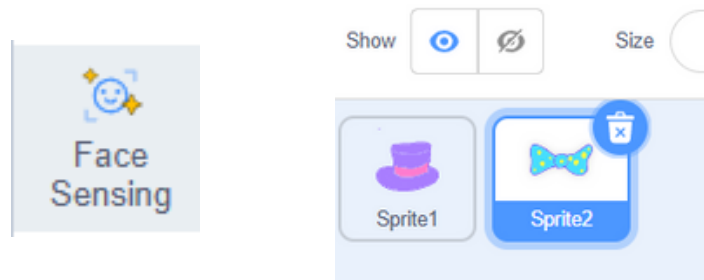
For more info watch this video:

<https://www.youtube.com/watch?v=uJVFvysUwa0>

Programming the AR face filter

Open the website <https://lab.scratch.mit.edu/face/> and click on the “try it out” button. You will then be directed to the Scratch environment that offers “face sensing”. We need these buttons to create our face filter.

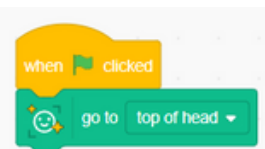
Make sure the AR element are uploaded and ready to be programmed. It will look like this.



Step 1: In the left column, click on the yellow tab with the word “event.” Next, select this block. This is the start block of your program. When you click on the green flag, which you can find in the left corner above your camera, your program will start

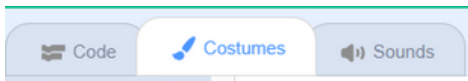


Step 2: Now click on “face sensing” in the left column and add this block to your program. Then select the desired body part. then it recognizes the body part and places the AR element at the desired location.



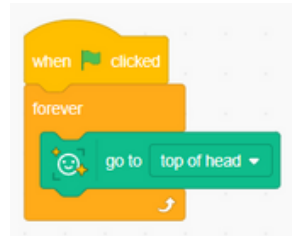
Preparation and teacher notes

If the size or placement of your AR element is not to your liking and you want to adjust it, select the AR element you wish to change and click on "Costumes." There, you can resize the element (make it bigger or smaller) or reposition it, for example, to place it higher on your head.



Programmeren van de AR-face filter 60'

Step 3: When you test the program you will notice that the AR element is placed on the desired body part only when pressing the green flag. We want this to follow the body part on the camera. Therefore, we add this block. This block ensures that face sensing block remains active. You can find this block under "control" on the left side.



Step 4: When you stand close or far away from the camera you will notice that the AR element does not change with the size of your head. To change this we add this block to the program. This block causes the AR element to increase in size when you are close to the camera and decrease in size when you are further away from the camera.



Step 5: Repeat these steps for the other AR elements. Add decoration or a painting frame to your living artwork.

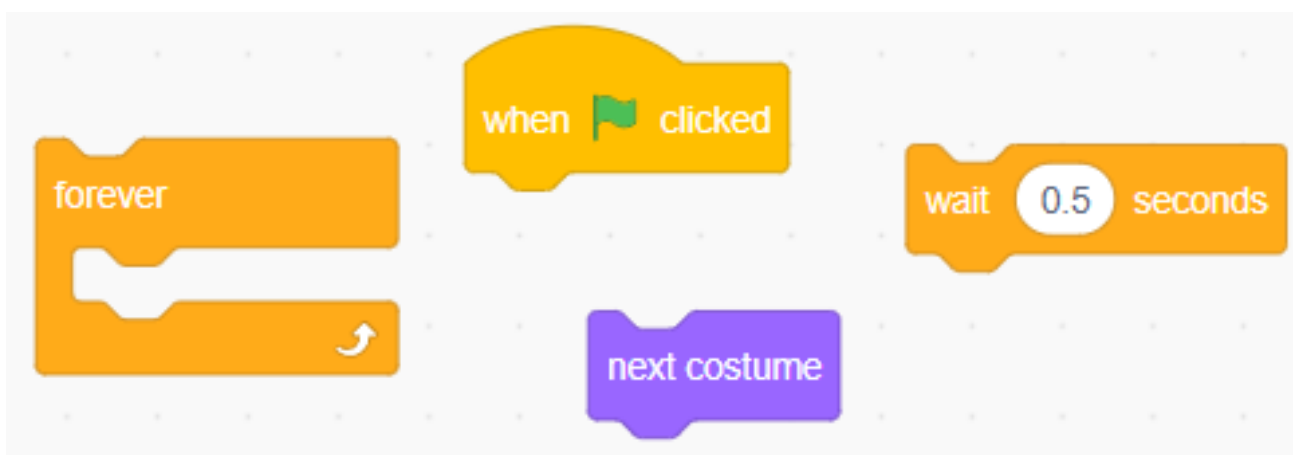
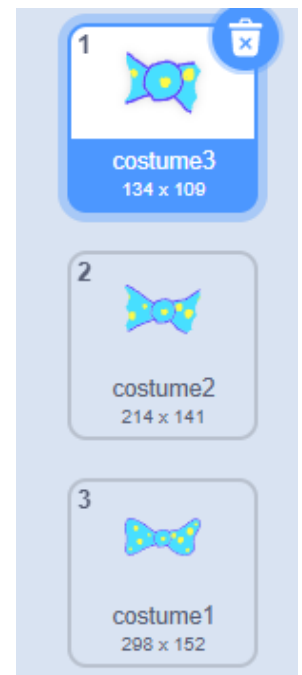
Step 6: Finally, hand out the inspiration sheets. Then let the children work independently to add additional elements to their life painting.

Can you animate your AR element?

Go to “costume” of the AR element you would like to animate. duplicate this several times. Then change something about each costume each time.

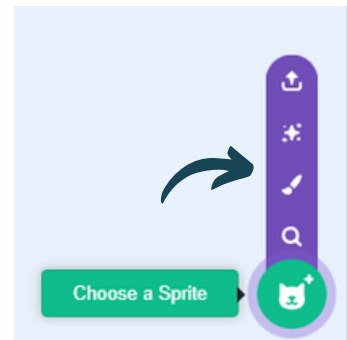
Example: Remove part of the AR element piece by piece. Then program your AR element so that a bit more of it appears on your screen.

Can you use these programming blocks to annimate your AR element?

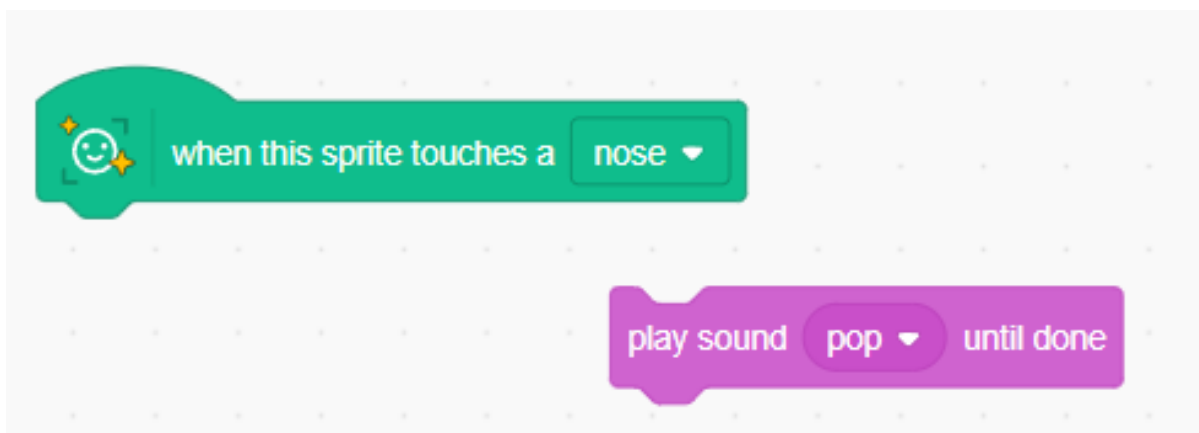


Can you make a sound with your nose?

Create a new figure by clicking on the brush then an empty sheet will open. On this sheet you can design a button. Then place the button on your camera image. Program the button so that when you touch it with your nose plays a sound.



Can you use these programming blocks to make your nose make a sound?

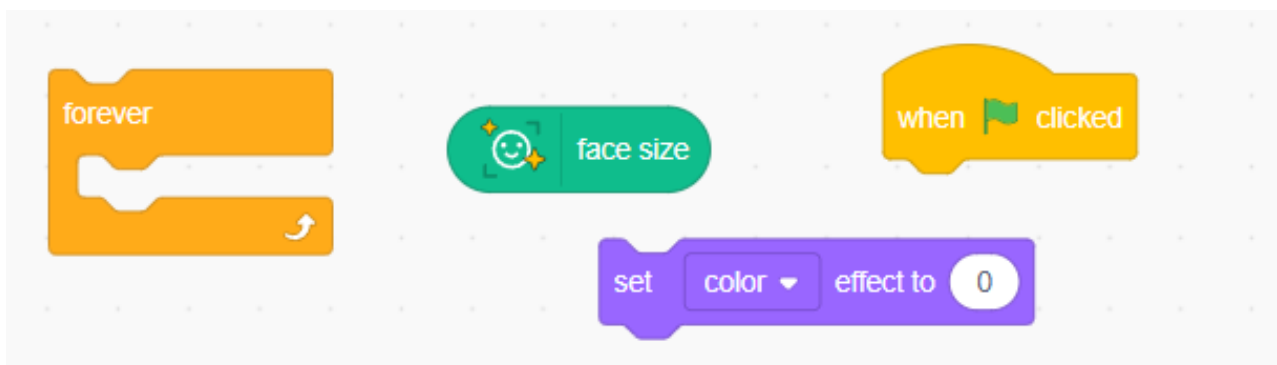


Can you make your AR element change color by making your head bigger or smaller?

Choose an AR element that you would like to change color. When your head gets smaller or larger, the color of your AR element changes. In this way your AR element can have all the colors of the rainbow.



Can you use these programming blocks to make your AR element change color by making your head bigger or smaller?



Can you make your painting move with your head?

Have you provided additional decoration and would you like to move it now? Select an element and program it with the blocks below.

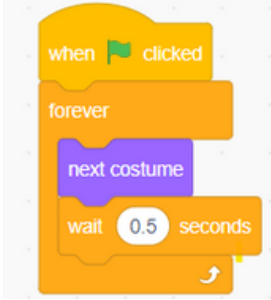
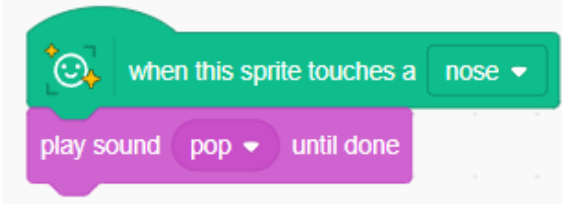
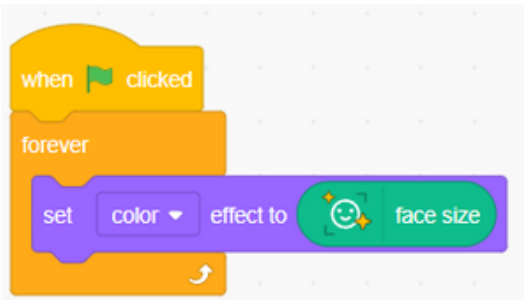
For example, this could be a flower standing at the edge of your painting. The more you tilt your head to the left or right, the harder the flower turns.



Can you use these programming blocks to make your decoration move with your head?



Program solution

Task	Solution
Can you animate your AR element?	 <pre> when green flag clicked forever loop next costume wait 0.5 seconds </pre>
Can you make a sound with your nose?	 <pre> when this sprite touches a nose play sound 'pop' until done </pre>
Can you make your AR element change color by making your head bigger or smaller?	 <pre> when green flag clicked forever loop set color effect to face size </pre>
Can you make your painting move with your head?	 <pre> when green flag clicked forever loop turn face tilt degrees </pre>

